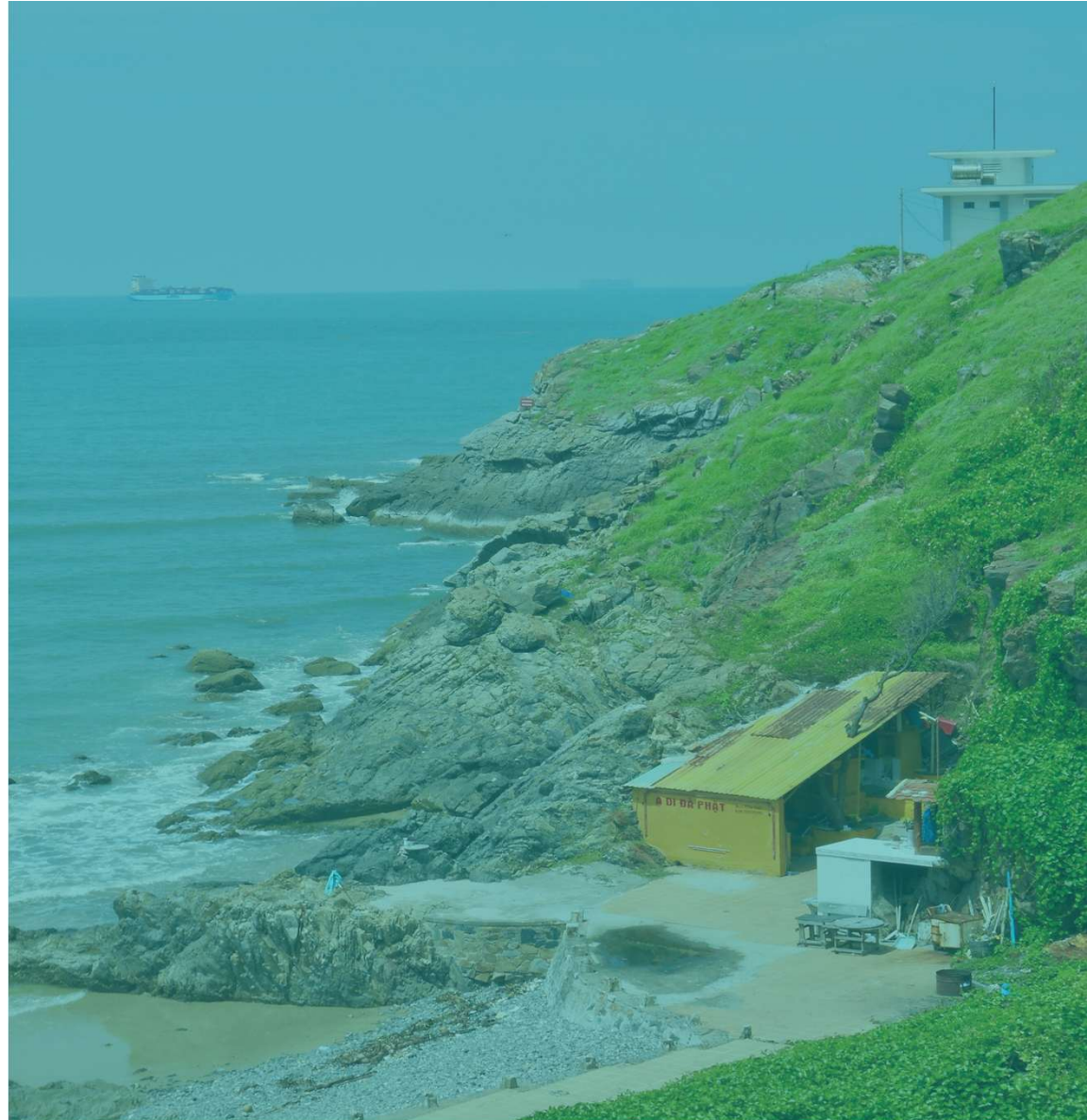


LING 620 Final Project

April 9, 2026

Revisiting Relative Clause Attachment by L1 Vietnamese-L2 English Bilinguals: Effects of Prosody and Proficiency

Presented by Clara Tran



Background

- Consider this sentence in English:

Jimmy met the brother of the engineer who has a smart dog.
Noun Phrase 1 Noun Phrase 2 Relative Clause

- Focus on attachment ambiguity: high attachment (HA) and low attachment (LA)
- Cross-linguistic mismatch in native speakers' default preference: English (LA) vs. Spanish/French (HA) (Fodor 2002; Cuetos & Mitchell 1988)
- Default attachment may reflect implicit prosody (Fodor 1998, 2002)
 - Readers assign a default prosodic contour during silent reading, which guides syntactic parsing and interpretation.
- Auditory stimuli may yield more consistent interpretations.

- Prosodic phrasing affects attachment:
 - Break after NP1 → LA preferred: NP2 + RC in same Intonational Phrase (IP)
 - Jimmy met [the brother] // [of the engineer who has a smart dog].
 - Break after NP2 → HA preferred: RC in separate IP
 - Jimmy met [the brother of the engineer] // [who has a smart dog].
- L2 learners' interpretation patterns may vary (transfer from L1, no preference, native-like)
- L2 learners can use prosodic cues to resolve ambiguity. (e.g., Dekydtspotter et al. 2008, Goad et al. 2021, Jun 2003)
- However, most studies focus on L1-L2 pairs with similar ambiguous structures.
- Vietnamese RCs are more frequent in formal registers, influenced by contact with European languages. (Ngo 2021)
- RCs following a complex NP (equivalent to 'NP1 of NP2' in English) are marginal although possible in Vietnamese → Such ambiguity is infrequent in their L1.

Research Questions

RQ1. Prosody:

Does prosodic condition (breaks) influence English RC attachment responses (high vs. low attachment) between groups (native speakers vs. learners)?

RQ2. Proficiency:

Does learners' proficiency (cloze score) modulate sensitivity to prosodic cues?

RQ3. Processing:

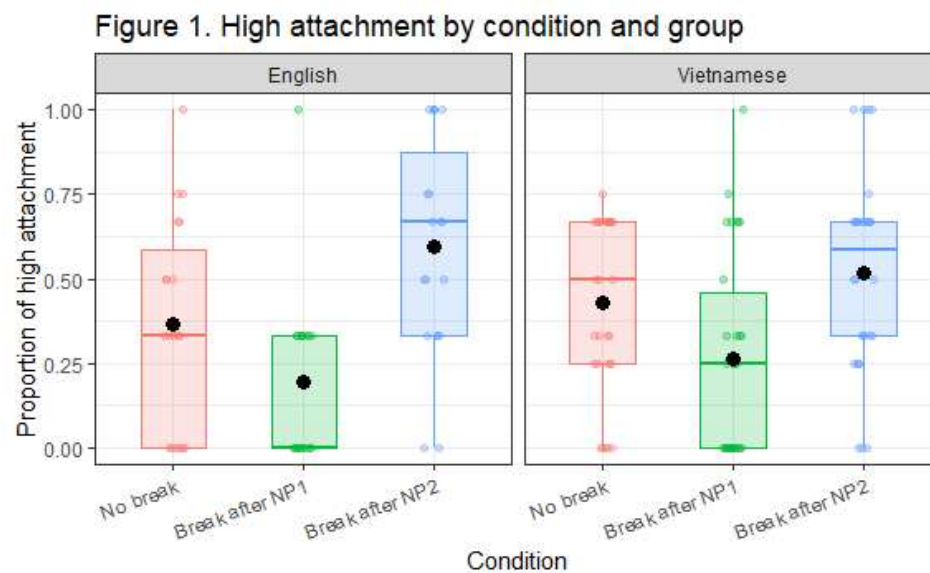
Do prosodic condition and proficiency affect reaction times (RTs)?

Dataset

- 490 observations of attachment response, corresponding to 10 different sentences with ambiguous RC attachment and 49 participants (30 Vietnamese learners of English and 19 native English speakers).
- The dataframe contains columns corresponding to several variables:
 - *Participant-level:*
 - `group`: English vs. Vietnamese)
 - `cloze`: proficiency score; only defined to L2 learners $\rightarrow$ `cloze_z` (used in learner model)
 - *Item-level:*
 - `condition`: 3 levels (`no_break`, `break_after_NP1`, `break_after_NP2`)
 - *Responses:*
 - `attachment_HA`: 1/0 for high/low attachment responses
 - `reactionTime` $\rightarrow$ `logRT` $\rightarrow$ `RT_z_part` (within-participant standardized)

EDA

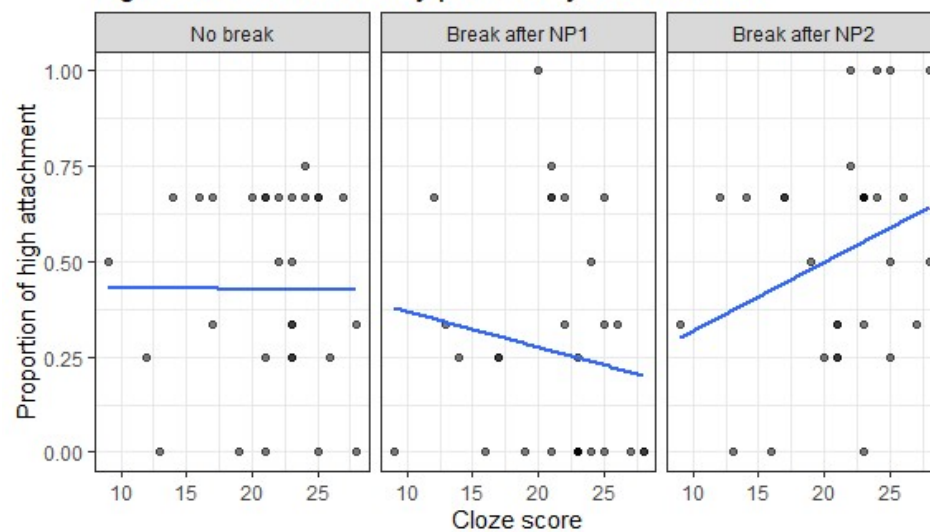
- HA by condition and group
 - Effect of prosody across both groups
 - Break after NP1: low HA proportion → LA preference
 - Break after NP2: high HA proportion
 - Learners show weaker separation between conditions.



EDA

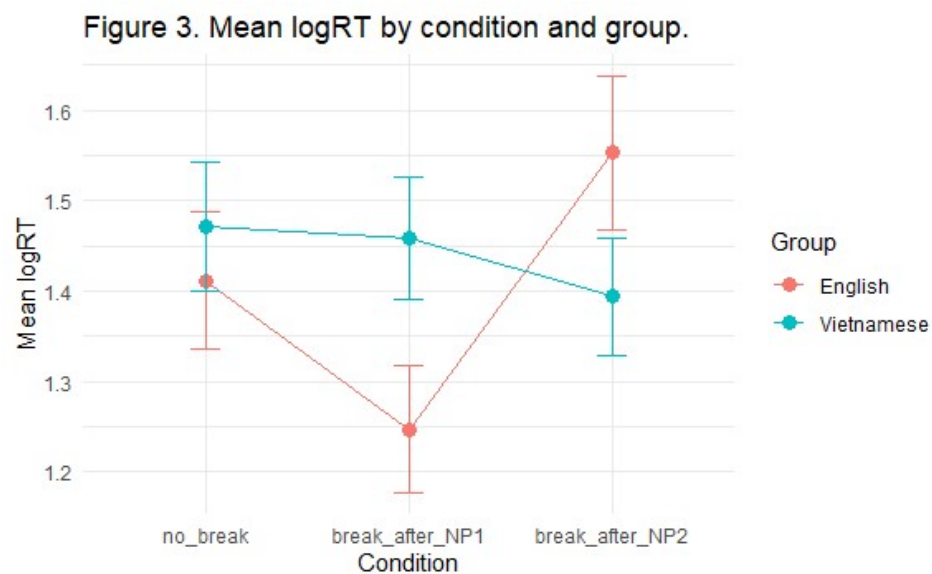
- Learners' HA by proficiency and condition
 - No clear effect of proficiency overall in *No break* condition
 - Weak trends in *Break after NP1* (slightly negative) and *Break after NP2* (slightly positive) conditions.
 - High variability across participants

Figure 2. Learners' HA by proficiency and condition.



EDA

- Processing time: Mean logRT by condition and group
 - No consistent effect of prosody on logRT: patterns differ across groups.
 - English: more variation across conditions
 - faster in NP1, slower in NP2
 - Vietnamese: subtle change across conditions
 - (surprisingly) faster in NP2 compared to the English group



Models

Model 1: Attachment between groups (GLMM)

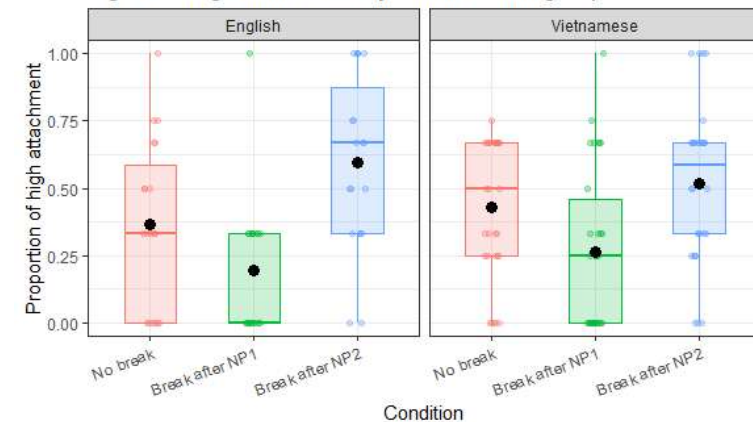
$\text{attachment_HA} \sim \text{condition} * \text{group} + (1 | \text{participant}) + (1 | \text{item})$

- Strong prosody effects
 - Break after NP1* $\rightarrow$ fewer HA
 - $(\beta = -1.05, p = .016)$
 - Break after NP2* $\rightarrow$ more HA
 - $(\beta = 1.03, p < .01)$
- Group effect: not significant
- No significant interaction between condition and group

	attachment_HA			
Predictors	Log-Odds	p		
(Intercept)	-0.60	0.100	Random Effects	
condition	-1.05	0.016	σ^2	3.29
[break_after_NP1]			τ_{00} participant	0.39
condition	1.03	0.010	τ_{00} item	0.32
[break_after_NP2]			ICC	0.18
group [Vietnamese]	0.23	0.557	$N_{\text{participant}}$	49
condition	0.21	0.696	N_{item}	10
[break_after_NP1] $\times$ group			Observations	490
[Vietnamese]			Marginal R^2 / Conditional R^2	0.103 / 0.262
condition	-0.58	0.250		
[break_after_NP2] $\times$ group				
[Vietnamese]				

Figure 1. High attachment by condition and group

Figure 1. High attachment by condition and group



Models

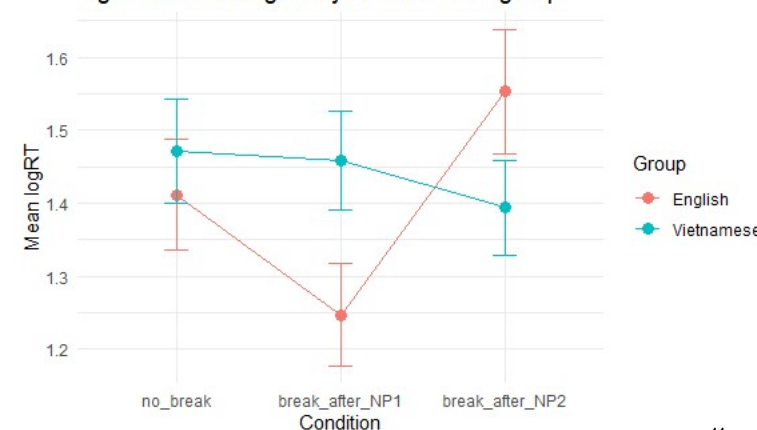
Model 3: Reaction Time (LMM)

```
log_RT ~ condition * group + (1 | participant)
+ (1 | item)
```

- Prosody effects
 - *Break after NP1* → faster RT (marginal)
 - ($\beta = -0.17$, $p = .097$)
 - *Break after NP2* → slower
 - ($\beta = 0.15$, $p = .14$)
- Group effect: not significant
- Marginal interaction (NP2 x Vietnamese):
 - Vietnamese participants may be less slowed down in NP2
 - ($\beta = -0.22$, $p = .093$)
- Random effects: substantial participant variability but minimal item variability

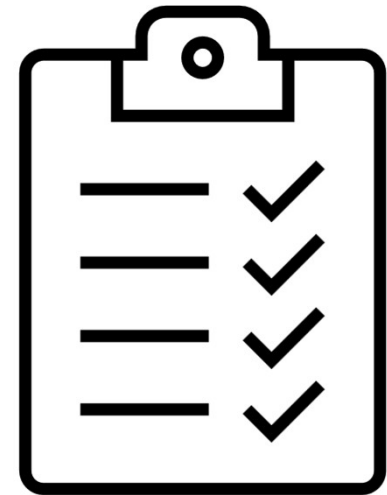
logRT				
Predictors	Estimates	p		
(Intercept)	1.41	<0.001	Random Effects σ^2 0.31 τ_{00} participant 0.11 τ_{00} item 0.00 ICC 0.27 $N_{\text{participant}}$ 49 N_{item} 10 Observations 487 Marginal R^2 / Conditional R^2 0.017 / 0.282	
condition [break_after_NP1]	-0.17	0.097		
condition [break_after_NP2]	0.15	0.141		
group [Vietnamese]	0.06	0.676		
condition [break_after_NP1] × group [Vietnamese]	0.15	0.235		
condition [break_after_NP2] × group [Vietnamese]	-0.22	0.093		

Figure 3. Mean logRT by condition and group.



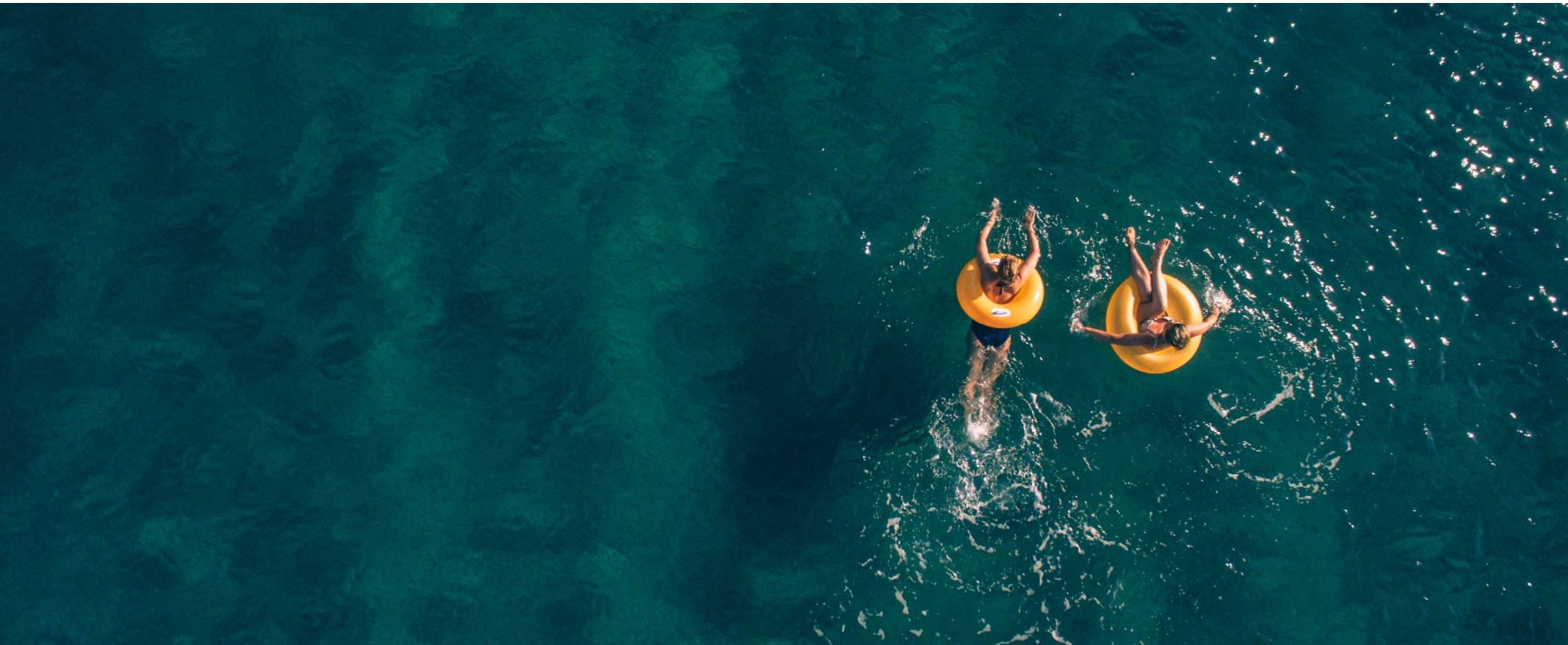
Next steps to consider

- ☐ Random slopes for *condition* (participant/item)
- ☐ Compare models (e.g., random slopes, interactions)
- ☐ Categorical measure of proficiency (e.g., Intermediate vs. Advanced)
- ☐ Use within-participant standardized RT (RT_{z_part}) for RT analysis
- ☐ Maybe effect of item ordering?
- ☐ Compare with previous findings (e.g., proficiency not significant)
- ☐ Relationship between interpretation and processing time



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Q & A

THANK YOU! XIN CẢM ƠN!